Year 13 Further Maths - Continuous random variables

Similar questions - CDFs, expectation and transformations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The continuous random variable T has cumulative distribution function $\mathrm{F}(t) = \begin{cases} 0 & t < 0 \\ 1 - \mathrm{e}^{-0.4t} & t \geqslant 0. \end{cases}$

(a) Find the cumulative distribution function of $3T$. **[3]**

(b) Show that, for constant k, $\mathrm{E}\left(\mathrm{e}^{kT}\right) = \dfrac{0.4}{0.4-k}$, and state with a reason the range of values of k for which this holds. **[7]**

(c) T is the time before a certain event occurs. Show that the probability that no event occurs between time 0 and time θ is the same as the probability that a random variable with distribution $\mathrm{Po}(\lambda)$ equals 0, for a value of λ which you should state in terms of θ. **[4]**

2 The continuous random variable X has probability density function $\mathrm{f}(x) = \begin{cases} \dfrac{k}{x^n} & x \geqslant 1 \\ 0 & \text{otherwise,} \end{cases}$ where $n > 1$ is an integer.

(a) Find k in terms of n. **[3]**

(b) When $n = 4$, find the cumulative distribution function of X and hence determine $\mathrm{P}(X > 6 \,|\, X > 3)$. **[6]**

(c) Find the values of n for which $\mathrm{Var}(X)$ is not defined. **[4]**

3 The continuous random variable Y is uniformly distributed on $[0, 2]$.

(a) It is given that $\mathrm{E}[a\cos(aY)] = 0.4$, where a is a constant between 0 and 1 and aY is measured in radians. Determine the exact value of a. **[5]**

(b) Determine the 60th percentile of Y^2. **[4]**

4 The continuous random variable X has probability density function $\mathrm{f}(x) = kx^n$ for $0 \leqslant x \leqslant 1$ (and 0 otherwise), where k is a constant and $n > 0$. It is given that the median of X is 0.8.

(a) Show that $n = \dfrac{\ln 0.5}{\ln 0.8} - 1$ and find n to 4 s.f. **[5]**

(b) Twenty independent observations of X are made. Find the expected number of observations that are less than 0.9. **[3]**

5 The continuous random variable X has a uniform distribution on the interval $[-\pi, \pi]$. The random variable Y is defined by $Y = \sin X$. Determine the cumulative distribution function of Y. **[7]**

Question 1

(a) $\mathrm{P}(3T \leqslant y) = \mathrm{P}\left(T \leqslant \frac{y}{3}\right) = 1 - \mathrm{e}^{-0.4y/3}$ for $y \geqslant 0$ (and 0 for $y < 0$)

(b) $\mathrm{f}(t) = 0.4\mathrm{e}^{-0.4t}$. $\mathrm{E}(\mathrm{e}^{kT}) = \int_0^\infty 0.4\mathrm{e}^{(k-0.4)t}\mathrm{d}t = \left[\dfrac{0.4}{k-0.4}\mathrm{e}^{(k-0.4)t}\right]_0^\infty = \dfrac{0.4}{0.4-k}$ provided $k < 0.4$ (otherwise the integral diverges)

(c) $\mathrm{P}(T > \theta) = \mathrm{e}^{-0.4\theta}$ and $\mathrm{P}(\mathrm{Po}(\lambda) = 0) = \mathrm{e}^{-\lambda}$; these are equal for $\lambda = 0.4\theta$

Question 2

(a) $\int_1^\infty kx^{-n}\mathrm{d}x = \dfrac{k}{n-1} = 1 \Rightarrow k = n-1$

(b) $k = 3$: $\mathrm{F}(x) = \int_1^x 3u^{-4}\mathrm{d}u = 1 - x^{-3}$ for $x \geqslant 1$. $\mathrm{P}(X > 6 \,|\, X > 3) = \dfrac{6^{-3}}{3^{-3}} = \dfrac{1}{8}$

(c) $\mathrm{E}(X^2) = \int_1^\infty (n-1)x^{2-n}\mathrm{d}x$ converges only if $n - 2 > 1$, i.e. $n > 3$. The variance is not defined for $n = 2$ and $n = 3$

Question 3

(a) $\mathrm{E}[a\cos(aY)] = \frac{1}{2}\int_0^2 a\cos(ay)\mathrm{d}y = \frac{1}{2}\sin 2a = 0.4$, so $\sin 2a = 0.8$. With $0 < a < 1$, $0 < 2a < 2$ so $2a = \sin^{-1} 0.8$ (the other solution $\pi - 0.927 = 2.21 > 2$ is excluded). $a = \frac{1}{2}\sin^{-1} 0.8 = \tan^{-1}\frac{1}{2} = 0.464$

(b) $\mathrm{P}(Y^2 \leqslant q) = \mathrm{P}(Y \leqslant \sqrt{q}) = \frac{\sqrt{q}}{2} = 0.6$, $q = 1.44$

Question 4

(a) $k = n+1$, $\mathrm{F}(x) = x^{n+1}$; median: $0.8^{n+1} = 0.5$, so $n+1 = \dfrac{\ln 0.5}{\ln 0.8}$, $n = \dfrac{\ln 0.5}{\ln 0.8} - 1 = 2.106$

(b) $\mathrm{P}(X < 0.9) = 0.9^{n+1} = 0.9^{3.106} = 0.721$; expected number $= 20 \times 0.721 = 14.4$

Question 5

For $-1 \leqslant y \leqslant 1$, let $\alpha = \sin^{-1} y$ $(-\frac{\pi}{2} \leqslant \alpha \leqslant \frac{\pi}{2})$. If $y \geqslant 0$, $\sin x \leqslant y$ for $x \in [-\pi, \alpha] \cup [\pi - \alpha, \pi]$, of total length $\pi + 2\alpha$; if $y < 0$ it holds for $x \in [-\pi - \alpha, \alpha]$, also of length $\pi + 2\alpha$. So $\mathrm{P}(\sin X \leqslant y) = \dfrac{\pi + 2\alpha}{2\pi}$. So $\mathrm{F}(y) =$

$$\begin{cases} 0 & y < -1 \\ \frac{1}{2} + \frac{1}{\pi}\sin^{-1} y & -1 \leqslant y \leqslant 1 \\ 1 & y > 1 \end{cases}$$